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Target Run-up and Cross-border M&A: Information Leakage and the Effect of Country-level Development

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Preface

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Executive Summary

This study investigates the extent of information leakage, proxied by target stock price run-up, in cross-border versus domestic merger and acquisition (M&A) transactions, and explores the role of country-level development.

Information leakage is when outsiders have knowledge of sensitive information prior to its public announcement, and, through reviewing existing literature on the topic, this study shows how target run-up can be used as a measure for this in the context of M&A deals. Literature shows that information leakage is greater when information asymmetries are more prevalent, thus leading to the hypothesis that cross-border deals have higher information leakage. This is tested using a multivariate regression framework, finding that a cross-border deal will result in run-up approximately 2.5% higher than an identical domestic transaction.

Two arguments are made regarding the role of development on information leakage. First, the income argument: that those in poorer countries have more to gain from informed trading, and thus greater incentives. And second, the environment argument: poorer countries have weaker institutions and less developed financial markets, making informed trading easier and information leaks more likely. Testing this through a linear regression framework for both acquiror and target nations, as well as the relative development, yielded insignificant results, suggesting that overall development does not play a significant role. However, an investigation into non-linear relationships yielded significant results that warrant further future research.

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Introduction

Cross-border merger and acquisition (M&A) activity has grown significantly over recent years, constituting 35% of global M&A activity in the past ten years by volume¹. Such transactions provide the opportunity for firms to expand operations internationally and seek the best acquisition targets globally. Furthermore, at the country level, it is a form of foreign direct investment (FDI) that enables developing countries to attract the capital and resources of large multinational companies. Studies such as Wang and Wong (2009), Bertrand and Zuniga (2006), and Otchere and Oldford (2018) have shown that receiving M&As can promote economic growth, R&D, innovation and financial market development for emerging economies.

Information leakages and target run-up are relevant to both acquiror and target firms in M&A transactions. Such leakages can put deal completion at risk (Ngo & Susnjara, 2016) or be costly to the acquiring firm Schwert (1996), implying that, if management recognises attributes that will result in higher information leakage, they will be less inclined to make an investment. Understanding the causes and patterns in such leakages can help acquirors choose suitable targets and is also important for regulatory and government bodies wishing to promote fair activities and economic development, thus especially pertinent to authorities in developing countries.

It is thus important for firms to understand the risks of information leakages when engaging in a transaction and determining which targets, countries and markets could put deals at risk. As such, minimising information leakages and target run-up can incentivise cross-border transactions, and benefit host countries. This study focuses on understanding cross-border M&A transactions and information leakage, and whether leakage is more prevalent in cross-border transactions than domestic transactions. Furthermore, a deeper look is taken into possible drivers of this, and whether country-level development influences leakage prior to M&A announcements.

Information leakage is most commonly identified through target stock price pre-announcement run-up. This occurs as those with inside information trade upon their knowledge of the upcoming deal, and thus the target firm's stock price exhibits abnormal returns in the period prior to the transaction's announcement. Of course, this is not a pure effect. Other factors also influence the

¹ The Harvard Law School Forum on Corporate Governance. (2023, January). *Cross-Border M&A – 2023 checklist for Successful acquisitions in the U.S.*

target stock price, for example a young company with new technology gaining traction may naturally attract investors and a higher stock price, whilst on the other hand attract M&A attention from incumbent firms wishing to acquire this technology. Nonetheless, this has thus far been used as an affective measure of information leakage, and using statistical processes this relationship can be analysed. Furthermore, research such as Schwert (1996) has shown that insider trading and information leakage prior to announcements does occur and can be shown through target run-up.

Brigida and Madura (2012) look at what causes stock price run-up prior to acquisitions, and the characteristics of transactions that exhibit higher run-up. Their main findings include that smaller target companies, especially in the technology sector, experience higher run-up, whilst friendly bids funded mostly through borrowing also result in higher run-up. They also examine the effect of Sarbanes-Oxley, which was US regulation that placed more onus on managers and stricter rules surrounding reporting of insider trades, and find that this reduced run-up, which is evidence suggesting run-up is possibly caused by insider trading that was then curbed by new regulations. The authors concluded that targets with greater levels of information asymmetry experience higher pre-announcement run-ups, and that this also depends on the regulatory environment.

Similarly, Tang and Xu (2016) investigate the causes behind pre-announcement target stock price run-up, with a greater focus on insider trading. With regards to international transactions, the authors hypothesise that cross-border deals have a lower impact on run-up because targets are less likely to expect the acquisition than in the case of domestic deals. Their empirical findings however suggest a significant slightly positive relationship between foreign bidders and run-up, although this is not further discussed and is not a focal point of the study.

Whilst both Brigida and Madura (2012) and Tang and Xu (2016) include variables to account for the effect of cross-border transactions, this is not the focus of either study. Furthermore, both use data on transactions where the target is from the US, thus limiting the application of the findings to other markets. Through using data from a wider range of markets, this study provides a more complete analysis of cross-border transactions and information leakage. The first research focus in this study aims to answer the question:

Is information leakage more prevalent in cross-border M&A transactions than for domestic deals?

Studies such as Xie, Reddy and Liang (2017) and Deng and Yang (2015) focus on country or regulatory development and the level of M&A activity, or the associated premiums, providing analysis on the determinants of the level and success rate of cross-border M&A. With respect to information leakage, there is a smaller body of work that focuses on a development angle. Feito-Ruiz and Menendez-Requejo (2011) investigates legal environments and cross-border M&A activity but investigates the returns for the bidder firm. The study considers pre-announcement acquiror price run-up for different levels of legal and institutional development, finding that stronger legal and institutional environments benefit acquiror shareholders, because they place a higher valuation on the M&A deal. By taking a broader approach to country-level development through using common macroeconomic indicators, this study provides evidence as to whether broader country characteristics play a role in information leakage. Thus, the second research question is:

Does the level of development of target and acquiror countries have an effect on the degree of information leakage in cross-border M&A transactions?

Using a sample of international cross-border and domestic transactions and the cumulative abnormal return (CAR) of the target firm in each transaction, this study uses a multivariate regression framework to determine the relationship between cross-border transactions and information leakage. Country-level development is measured by GDP and supported by measures of regulation, corruption and the rule of law to examine the effect of development on this relationship.

The research expectation is that when there is a larger difference in development between two countries involved in a transaction, there is a greater incentive and opportunity to leak information, thus information leakage will be more prevalent.

By investigating cross-border transactions, country-level variables that reflect characteristics of respective populations and environments can be used to analyse the factors affecting information leakage and target stock price run-up. Using a range of countries that host both acquiring and target firms gives a novel perspective for this type of analysis.

Literature Review

Information leakage occurs when individuals obtain information prior to when it is available to the market². This can be a result of information being stolen for example through hacking, or simply employees unwittingly prematurely sharing private information. In this study, information leakage includes insider trading, or more broadly informed trading which includes situations when market participants trade on information sourced from within the company that is not yet publicly available.

To study information leakage, it is most insightful to look at target firm stock price run-up. There is conflicting research on whether M&A announcements are value creating or destructive for acquiring firms, but there is a consensus that target-firm shareholders positively value such announcements (Feito-Ruiz et al., 2011). As such, those who are prematurely informed regarding a transaction will be able to profit through purchasing target stock, as this will most likely be valued higher as a result of the transaction. In the event of information leakage, we should therefore observe abnormal trading and run-up of the target's stock.

Dutodoir, Vagenas, Verwijmeren and Wu (2021) propose two non-mutually exclusive hypotheses which could explain target run-ups, namely “Deal Anticipation” and “Insider Trading”, whereby the first involves legal trading practices and the latter is illegal in most countries (Bhattacharya and Daouk, 2002). Dutodoir et al (2021) attempt to explain the decline in target run-up between 1985 and 2018, and find that lower levels of insider trading are associated with lower run-up, thus implying insider trading is a primary cause of target run-up. Although the deal anticipation hypothesis was not found to be instrumental in this study, it is still relevant as part of information leakage more broadly for our purposes. Information leakage in this paper combines insider and informed trading as well as any other form through which investors become aware of internal information. Deal anticipation consists of rumours and close attention to companies. Thereby both the deal anticipation and insider trading hypotheses are constituents of information leakage, and thus the two key explanations for target run-up from Dutodoir et al (2021) and the literature more

² Law, J. (2008). A Dictionary of Finance and Banking. In *Oxford University Press eBooks*. <https://doi.org/10.1093/acref/9780199229741.001.0001>

broadly are indicative of information leakage. Henceforth, I will refer to literature on both information leakage and run-up equally, with both taken to indicate information leakage.

Information leakage is closely related to market efficiency. The strong form of market efficiency states that all information pertaining to a stock is incorporated into its price, including public, private, and past price information. Many studies prove that this is true to varying extents, and none more prominently than the papers from Eugene Fama. Fama (1970) supports the weak form of market efficiency, namely that historical price patterns are incorporated into current prices, and also implying that private information is not incorporated.

In their literature review on market efficiency, Dimson and Mussavian (1998) conclude that insider trading which results in excess returns violates the strong form of the efficient market hypothesis. From this, one can draw that if the market is strongly efficient, then insider trading is not possible, and, as this is one of its major components, information leakage may also not occur. So, the efficiency of a market is an important determinant of information leakage, which is especially pertinent in the later discussion of development.

Schwert (1996) presents one of the first articles investigating pre-announcement run-ups with a focus on M&A transactions. This article shows that run-up is typically a cost to the bidder, as there is little substitution between the premium paid and run-up, meaning that run-up is a cost in addition to the usual premium paid during a transaction. The author takes this run-up as evidence of informed trading, however questions whether this is indeed a bad thing, or instead facilitating more efficient markets.

More recently, Brigida and Madura (2012) and Tang and Xu (2016) research the determinants of stock price run-up. Both papers use data from US target firms, with the majority of deals being between US firms. Brigida and Madura (2012) find that higher information asymmetries result in greater target run-up and information leakage. Included in the sources of information asymmetries are firm size, with smaller targets resulting in greater information leakage, and also that foreign bidders are a characteristic of transactions with higher target stock price run-up. Tang and Xu (2016) put a greater focus on insider trading. Their main findings include that reported insider trading does not influence run-up, but higher media attention on insider trading and institutional ownership reduce pre-announcement abnormal returns. In line with this, they find that a measure

showing the probability of informed trading is positively related to price run-up. With regards to international transactions, the authors hypothesise that cross-border deals have a lower impact on run-up because targets are less likely to expect the acquisition than in the case of domestic deals. Their empirical findings however suggest a significant slightly positive relationship between foreign bidders and run-up, although this is not further discussed and is not a focal point of the study.

In recent years, there has been much more attention devoted to international M&A activity. One such study is Ngo and Susnjara (2016), which primarily studies hostility and the probability of deal completion for international transactions. Most relevant to this study though is their investigation into information leakage as a moderating effect. Using data from bidders and targets from across the world, they find that prevention of information leakage is crucial in ensuring the success of M&A deals, emphasizing the pertinence of understanding this phenomenon and its causes.

Cross-border transactions are more complex as they typically involve contrasting corporate cultures, institutional environments and more parties involved to help facilitate the transaction and bridge these differences. This complexity allows for more opportunities for mistakes resulting in information leakage. However, for insiders in a cross-border transaction, it may be more difficult to trade on the expectation of an event due to the company being listed on a different exchange, which would thus reduce the level of information leakage. The expectation of this study is that the former outweighs the latter, especially with the increased ease of trading internationally. The first hypothesis in this study is thus:

Hypothesis 1: Cross-border transactions result in greater information leakage than domestic deals.

The relationship between economic development and financial development is still a matter of discussion amongst academics. Early thought as far back as Smith (1776) and Schumpeter (1911) believed finance was an instrumental determinant of economic growth. In the mid- to late- 1900s, this opinion was contested, with papers such as Meier and Seers (1984), Lucas (1988) and Robinson (1952) suggesting that the finance industry grows to meet the needs of the real economy. Nowadays, evidence suggests that finance and especially strong financial institutions stimulate economic growth (Miller, 1998; Guru and Yedav, 2019). Regardless of which drives which, it is

widely observed that more economically developed countries feature more developed financial sectors, markets and institutions.

Rossi and Volpin (2004) find that in countries with greater shareholder protection there is more M&A activity. With regards to cross-border transactions, target firms are more often in country with less shareholder protection. Feito-Ruiz and Mene`ndez-Requejo (2011) explains this is because targets being acquired by a firm with superior shareholder protection can reduce its cost of capital through improving its institutional environment, although the benefits for the acquiror are not so clear.

To create a conceptual framework of how development influences information leakage, we can formulate two components that influence the decision to leak information: opportunity and incentive. Opportunity reflects the chance that an individual has to trade on inside information or share this with outsiders without being reprimanded. From a development perspective, this could be measured through looking at the strength of institutions and regulations within a country. Earlier discussions of Dutodir et al (2021) and Brigida and Madura (2012) have shown that information leakage is influenced by regulatory environment, suggesting that opportunity is a key determinant.

Incentive represents what an individual stands to gain by exploiting information leakage. This is harder to measure since we do not have access to the profits any individual makes from sharing or trading on inside information. However, to assist the information leakage and development framework, we can consider what individuals in different countries may stand to benefit from leaking information. For example, an individual with a lower income could be considered to have a greater incentive to leak information because they will put a greater weight on the income generated than someone who already has a higher income. At a country level, this can be proxied by looking at average income or wealth per person.

In answering the second research question as stated in the Introduction, the role of development is broken down into three hypotheses.

Firstly, the development of the acquiring country could matter, because this influences the likelihood of an information leakage due to the level of development of institutions and regulation (opportunity), and the incentive for holders of inside information to trade on this because of varying

levels of income. A less developed country with lower quality of and poorly enforced regulations allows for more insider trading and exploitation of inside information. Hence the hypothesis:

Hypothesis 2.1: Information leakage is greater for cross border M&A transactions where the acquiring company is from a less developed country.

Similarly, the development of the target country could matter for the same reasons. Employees of a target company in a less developed country will be less likely to be caught trading on inside information if institutions are also less developed. Additionally, as this is the country where the profitable trade on the target company can be made, hosting possibly the exchange and regulatory body governing the target firm, the reliability and development of these are even more important. Thus, the hypothesis:

Hypothesis 2.2: Information leakage is greater for cross border M&A transactions where the target company is from a less developed country.

Finally, the difference in development between the two countries may play a moderating role on the development of the acquiror. As a resident of a less developed country that hosts a company which is an M&A target of a firm from a more developed country, the expected premium and thus rise in stock price of the target may be higher, resulting in a greater incentive to trade on or leak inside information. This results in the final hypothesis:

Hypothesis 2.3: Information leakage is greater for M&A transactions where the acquiror's country is relatively more developed compared to the target's.

Testing each of these hypotheses will shed light on the ways in which development influences information leakage. Furthermore, by including both economic and institutional development measures, a conclusion can be made about the relative importance of opportunity and incentive in the context of information leakage.

Methodology and Data

Sample

The sample used consists of a panel dataset of 16,223 M&A transactions between 1978 and 2024 with targets in 50 countries sourced from LSEG Workspace. This is combined with stock price statistics from the WRDS Event Study tools. The limiting factor in terms of countries chosen is due to the data available from WRDS, whilst the time period is dictated by limits on LSEG Workspaces. Nonetheless, the sample is of sufficient size to effectively test the hypotheses and covers a broad range of countries with sufficient variation in development, and larger than most of the samples used in similar studies discussed in the Literature Review. Further country-level data is sourced from Worldbank.

Qualifying transactions are M&As that are flagged as Disclosed or Undisclosed Dollar Deal Value as defined under LSEG's M&A Types, with the percentage of shares acquired in the transaction at least 50.1%. Furthermore, in order to calculate run-up, only deals with a public target are included. Stake purchases (deals for a minority stake or consolidating a majority stake), Repurchases deal (target repurchasing its own shares), and Self-tender or recapitalisation deals are excluded. The former is excluded because this study is interested in companies newly acquiring control of another firm, and the remaining excluded categories do not bear the same informational asymmetries and competitive characteristics as the M&A transactions being studied.

Methodology

To analyse the relationship between cross-border deals and information leakage, as well as the impact of differences in country-level development, this study uses a multivariate regression framework. Similar methodologies are used extensively in the literature concerning run-up (Brigida & Madura, 2012; Ngo & Susnjara, 2016; Dutodoir et al., 2021). Target nation and year fixed effects are employed to improve the model fit. As demonstrated by Dutodoir et al. (2021), target run-up has changed over time, thus including year fixed effects removes issues from changing levels of average CAR over time and isolates the relationship of interest. Country fixed effects are used to control for unobserved, unmeasurable heterogeneity between countries, for example cultural differences and business traditions that are difficult to quantify. As with the majority of economic literature, robust standard errors are employed.

Dependent Variable

The level of information leakage is measured, as with the majority of existing literature, by the Cumulative Abnormal Return (CAR) for which Compustat's International Event Study and the WRDS U.S. Daily Event Study tools are used, following calculation specification similar to Schwert (1996) and Ngo and Susnjara (2016). The CAR in the period from 60 days to 1 day (-60, -1) before the announcement of the deal is used, however additional windows are used as robustness tests. Abnormal returns are calculated using the market-adjusted return model as shown below:

$$E(R_{i,t}) = R_{M,t}$$

$$AR_{i,t} = R_{i,t} - E(R_{i,t})$$

$$CAR_i(-60, -1) = \sum_{t=-60}^{t=-1} AR_{i,t}$$

Where:

$R_{M,t}$ is the market return, using the target country's market index.

$AR_{i,t}$ is the daily abnormal return for each of the days in the window (-60, -1) days prior to announcement.

$CAR_i(-60, -1)$ is the cumulative abnormal return for the same period, and the dependent variable in the regression. The literature uses a range of windows to measure run-up, with sixty days marking the longest considered. This window is chosen to maximise the capture of insider trading and information leakage, and thus used as the primary measure of run-up however alternative windows are also used, namely $CAR(-20, -1)$ and $CAR(-40, -1)$. Unless otherwise specified, CAR refers to $CAR(-60, -1)$ throughout this study.

Whilst many other factors affect CARs, the control variables used in the regression framework and furthermore the size and nature of the sample isolate the presence of information leakage causing any trends in CARs across the dataset. Further robustness tests are performed using Buy-hold Abnormal Returns, and the Cumulative Total Return, which are also sourced from the

aforementioned WRDS tools. The calculation of these variables is included in Appendix X. All measures of run-up are Winsorised at the 1% level.

Independent Variables

Firstly, this study is interested in the difference between CARs in domestic versus cross-border transactions. Thus, the first independent variable of interest is a dummy variable *Cross Border* which takes a value of 1 to indicate a cross-border transaction, and 0 for domestic deals. This is indicated in LSEG Workspaces as cross-border deal flag. The expectation is a positive relationship between *Cross Border* and *CAR*.

To test Hypotheses 2.1 and 2.2, country-level development variables are used for the acquiring and target countries respectively.

Targ_LogGDP and *Acq_LogGDP* are the logarithms of the gross domestic product for the target and acquiror nation respectively, measured in billions of current US dollars, thus accounting for changes in inflation. This captures the size and development of the respective economies as a whole. The natural logarithm is taken to account for the right skew in GDP. The expectation is that more developed countries will have lower levels of information leakage due to stronger institutional environments. However, the true relationship may be non-linear. It could be argued that, whilst for countries with high levels of development the institutional infrastructure is strong thus implying low levels of information leakage, countries with very low levels of development may also have low levels of information leakage due to limited access to capital markets. Initially, as development rises, individuals gain better access to capital markets and when the opportunity arises are able to make informed trades, until development is sufficiently high that regulators prevent inside trades. This is discussed further and modelled in the Results section.

Targ_GDPperCapita and *Acq_GDPperCapita* are respectively the target and acquiror nations' GDP per capita, measured in current US dollars, and aim to capture the individual wealth of residents of each country. This is used as a proxy for individual incomes, and thus shows the incentive aspect of information leakage as discussed in the Literature Review. The expectation is that, for lower individual levels of wealth (lower GDP per Capita) there is a greater incentive to trade on or leak inside information, and thus target run-up would be increasing in GDP per Capita.

For Hypothesis 2.3, the difference in development is measured as the ratio of acquiror development to target development, as shown below:

$$GDP_{ratio} = \frac{\log GDP_{Acquiror}}{\log GDP_{Target}}$$

$$GDP_{perCapitaRatio} = \frac{GDP_{perCapita}_{Acquiror}}{GDP_{perCapita}_{Target}}$$

These ratios are then separately interacted with the appropriate measure from the acquiror nation. As such, a large difference in GDPs will have a larger weighting when the acquiror is more developed, and mitigating contradictions with the hypotheses 2.1 and 2.2. Net National Income (*NetNI*), measured in billions of current US dollars, is used as an alternative measure to GDP, and included in robustness checks in the Appendix.

Transaction-level Controls

LogTotalAssets – the natural logarithm of the target firm’s total assets, measuring the size of the target firm. Studies such as Ambrose & Megginson (1992) and Song & Walkling (2000) suggest that smaller firms are more likely to be targets in M&A deals, which would imply an increase in deal anticipation, thus increasing run-up. The natural logarithm is taken to account for the right skew in Total Assets.

BooktoMarket – the target’s book-to-market ratio. Dutordoir et al. (2021) argue that a firm with a high book-to-market ratio is perceived as cheaper and may have underutilised opportunities that would be improved by a takeover, increasing shareholder value. Thus, a high book-to-market ratio would be associated with a greater likelihood of a takeover and result in higher anticipation and run-up. The book to market ratio is Winsorised at the 2% level to reduce the impact of outliers.

HighTech – a dummy variable indicating whether the target company is from a high-tech industry, as defined by LSEG Workspaces. High-tech industries are subject to more informational asymmetries, which allows insiders to better exploit their informational advantages, causing higher run-up (Brigida & Madura, 2012).

SharesAcquired – the portion of shares acquired in the transaction. A higher percentage of shares acquired could suggest more gains for target shareholders as more target shareholders are bought out. It could also suggest a smoother post-merger integration process as the acquiring firm has greater control of the firm, reducing post-merger inefficiencies and increasing the value generated by the merger. This would thus increase target run-up as a greater premium is anticipated.

TargetStockOptions – a dummy variable indicating when there are listed stock options traded on the target's stock. Literature such as Cao (1999) suggests that when derivatives are traded on a stock, the underlying price becomes more informationally efficient. It also gives insiders more opportunity to trade on inside information, which increases run-up (Brigida & Madura, 2012).

BorrowingsFinancing – a dummy variable taking a value of 1 when the transaction is financed through borrowing. By involving other organisations to assist in financing the transaction, there are more parties involved and a longer time period in which information may be leaked, resulting in higher run-up (Brigida & Madura, 2012). Furthermore, Acharya and Johnson (2007) suggest that, to an extent, information is leaked when investment banks participate in a deal as an underwriter and hedge their position, resulting in public trades and information leakage.

PercentageofCash – the portion of the deal which is paid for in cash by the bidder. An acquisition which is paid for in cash will require the bidder to build a large cash reserve. This may attract attention from insiders or observers of the company, and thus make it possible to anticipate the deal, which is shown to positively influence run-up by Brigida and Madura (2012).

Acq_Public – a dummy variable indicating whether the acquiror is publicly listed, as opposed to privately or government owned. Barger et al. (2008) show that public acquirors pay more than private bidders, which increases the anticipated value for target shareholders, thus increasing run-up.

PEBidder – a dummy variable indicating whether the bidder is a private equity firm. Brigida and Madura (2012) find that when the bidder is a private equity firm the target run-up is significantly lower due to the firm's expertise in transactions and ability to minimise leakage.

SameIndustry – a dummy indicating whether the two firms are from the same macro industry. Ngo and Susnjara (2016) suggests that, when two firms are more similar, there will be less need for

external consultation, thus minimising avenues for information leakage. However, a counter argument is that firms in the same industry will have a smoother, less costly post-merger integration process, generating more value and thus higher run-up.

Hostility – a dummy variable indicating whether the acquisition approach is hostile, giving value 1, or friendly, giving 0. Hostile bids have been shown to be associated with lower information leakage (Brigida and Madura, 2012) because the target's insiders will be less likely to expect and thus trade on the upcoming announcement.

Management – a dummy variable indicating whether management are involved in the acquisition. As Schwert (1996) finds that run-up is a cost to the bidder, if management are bidding for the firm, then they will want to avoid information leakage, thus reducing run-up. This expectation is also proven in Brigida and Madura (2012).

NumberofBidders – the number of bidders including the acquiring firm. As shown by Dutodir et al (2021), run-up is positively related to the number of bidders due to greater anticipation of the deal. Furthermore, more bidders would suggest more competition for the acquisition, resulting in a bidding war that drives up the acquisition price and premium, benefitting target shareholders and incentivising informed trading.

TotalAdvisors – the number of financial and legal advisors employed by both the bidder and the target. Similar to the motivation behind *BorrowingsFinancing*, involving more advisors gives more people inside information and thus increase the likelihood of information leakage (Brigida & Madura, 2012).

Toehold – the portion of the target that the bidder owns at announcement, taking values between 0 and 1. Following similar argumentation to *PercentageofCash*, an acquisition from a bidder who has already acquired a share of the target may be more easily anticipated (Betton et al., 2008).

Whilst *TotalAdvisors* and *Toehold* both have strong theoretical motivations, they are excluded from the models in this study due to the relatively low number of observations for both the sample as a whole, and more importantly for the cross border part of the sample, as shown in Tables 1 and 3

Country-level Controls

Country-level controls are sourced from the World Bank database and apply to the Target Nation. With the exception of *ListedCompanies* and *MarketCaptoGDP*, these are measured in units of the standard normal distribution, typically between -2.5 and 2.5.

ControlofCorruption – an estimate of the level of corruption in a country. A more corrupt country will be more likely to feature insider trading or information leakage as officials may be involved in insider-trading schemes as well. Furthermore, Weitzel and Berns (2006) find that targets in countries with high levels of corruption are more attractive, which could result in a higher premium being paid, increasing the incentives to make informed trades.

RegulatoryQuality – an estimate of the government's ability to implement policies to allow for growth in the private sector. Stronger, well-implemented regulations will make it harder to leak inside information, thus reducing run-up.

RuleofLaw – an estimate of confidence in and adherence to societal rules such as contract enforcement and property rights, but also crime and violence. As with regulations, abidance to laws is expected to reduce insider trading and information leakage (Madura, Ngo & Susnjara, 2014).

GovEffectiveness – an estimate of how effective the government is based on the quality of civil and public services, and its independence of political pressures. A more effective government will be better at implementing policies to curb insider trading resulting in lower run-up.

PoliticalStability – shows the perceptions of the likelihood of instability or politically-motivated events, including but not limited to terrorism. A more politically unstable country is expected to pay less attention to issues such as insider trading, resulting in higher levels of information leakage and higher run-up.

MarketCaptoGDP – the ratio of the sum of the market value of the listed companies in a country to its GDP. This represents the size of financial markets compared to the size of the country as a whole, and is a proxy for the level of capital market development (Li & Halebilan, 2021; Li, Arikan, Shenkar & Arikan, 2020). A country with more developed financial markets would have a higher ratio (La Porta, Lopez-de-silanes & Shleifer, 2006). A country with more financial development

would be expected to have more efficient financial markets, and thus exhibit fewer instances of abnormal returns, thus lower run-up.

ListedCompanies – the number of publicly traded companies in the country, shown as thousands of companies. More public companies would suggest more developed capital markets and greater financial development, which makes it harder to trade on inside information, reducing information leakage and run-up (Madura, Ngo & Susnjara, 2014).

Summary Statistics

Table 1 Full sample descriptive statistics

	N	Mean	Median	SD	Min	Max
CAR	16,223	0.0941	0.0644	0.2707	-0.6859	1.1225
CRET	16,223	0.1093	0.0667	0.3047	-0.6208	1.3846
BHAR	16,223	0.0817	0.0420	0.2917	-0.6171	1.3314
Cross Border	16,223	0.2566	0.0000	0.4368	0	1
LogTotalAssets	15,210	5.5817	5.4999	2.0371	-8.2983	15.1416
BooktoMarket	13,673	2.4268	1.5076	3.3699	-2.9803	18.3228
HighTech	16,223	0.3216	0.0000	0.4671	0	1
SharesAcquired	16,223	0.9060	1.0000	0.1600	0.5010	1
TargetStockOptions	16,223	0.3547	0.0000	0.4785	0	1
BorrowingsFinancing	16,223	0.1176	0.0000	0.3222	0	1
PercentageofCash	16,223	0.5255	0.6662	0.4754	0	1
Acq Public	16,223	0.5644	1.0000	0.4959	0	1
PEBidder	16,223	0.0518	0.0000	0.2217	0	1
SameIndustry	16,223	0.5731	1.0000	0.4946	0	1
Hostility	16,042	0.0221	0.0000	0.1471	0	1
Management	16,223	0.0311	0.0000	0.1737	0	1
NumberofBidders	16,223	1.0594	1.0000	0.2776	1	6
TotalAdvisors	7,701	6.7202	6.0000	2.9536	4	36
Toehold	2,229	0.2143	0.1999	0.1462	0	0.4990
Targ LogGDP	15,826	8.3358	8.5634	1.2961	1.2660	10.1441
Acq LogGDP	15,732	8.1795	8.5108	1.4772	-0.1172	10.1441
GDPratio	15,732	0.9864	1.0000	0.1439	-0.0143	4.5751
Targ GDPperCapita	15,826	37.3062	36.6102	17.044	0.3462	184.6398
Acq GDPperCapita	15,732	37.6412	36.8495	17.8524	0.3707	133.7118
GDPperCapitaRatio	15,732	1.2152	1.0000	2.4779	0.0122	87.9230
Targ LogNetNI	15,805	8.1488	8.3986	1.3261	0.9295	9.8829
Acq LogNetNI	15,609	8.0214	8.2938	1.4450	-0.2444	9.8829
NIratio	15,603	0.9902	1.0000	0.1442	-0.0480	5.1307
Targ NetNIperCapita	16,044	31.0404	31.1669	13.9031	0.3045	81.2756
Acq NetNIperCapita	15,609	30.7879	31.0659	14.0247	0.2686	82.5792
NIperCapitaRatio	15,598	1.1866	1.0000	2.3057	0.0124	85.0765
ControlofCorruption	13,520	1.3520	1.5367	0.6207	-1.1601	2.4541
RegulatoryQuality	13,520	1.3619	1.5325	0.4965	-1.2351	2.1811
RuleofLaw	13,521	1.3785	1.5628	0.5152	-1.5286	2.1248
GovEffectiveness	13,520	1.4185	1.5370	0.4675	-1.6840	2.3472
PoliticalStability	13,521	0.5619	0.6324	0.5428	-2.3760	1.7587
MarketCaptoGDP	15,791	1.0813	1.0810	0.6481	0.0735	17.7723
ListedCompanies	15,819	3.9593	4.1800	2.4701	0.0140	12.7300

Table 2 Correlations of firm- and deal- level variables

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) car	1.0000											
(2) CrossBorderDea~g	0.042* (0.000)	1.0000										
(3) LogTotalAssets	-0.090* (0.000)	-0.0010 (0.930)	1.0000									
(4) BooktoMarket	0.055* (0.000)	0.044* (0.000)	-0.084* (0.000)	1.0000								
(5) HighTech	0.032* (0.000)	0.055* (0.000)	-0.193* (0.000)	0.142* (0.000)	1.0000							
(6) SharesAcquired	0.020* (0.009)	-0.068* (0.000)	0.133* (0.000)	0.0120 (0.164)	0.079* (0.000)	1.0000						
(7) TargetStockOpt~s	0.026* (0.001)	-0.102* (0.000)	0.168* (0.000)	0.071* (0.000)	0.157* (0.000)	0.367* (0.000)	1.0000					
(8) BorrowingsFina~g	0.031* (0.000)	0.044* (0.000)	0.088* (0.000)	0.022* (0.009)	0.0090 (0.266)	0.062* (0.000)	0.061* (0.000)	1.0000				
(9) PercentageofCash	0.027* (0.001)	0.206* (0.000)	-0.057* (0.000)	-0.0070 (0.396)	0.097* (0.000)	-0.024* (0.003)	0.085* (0.000)	0.235* (0.000)	1.0000			
(10) Acq_Public	0.017* (0.033)	-0.148* (0.000)	0.127* (0.000)	0.034* (0.000)	0.047* (0.000)	0.263* (0.000)	0.223* (0.000)	-0.060* (0.000)	-0.291* (0.000)	1.0000		
(11) PEBidder	-0.036* (0.000)	0.043* (0.000)	0.022* (0.007)	0.0160 (0.063)	0.045* (0.000)	0.031* (0.000)	0.031* (0.000)	0.058* (0.000)	0.170* (0.000)	-0.236* (0.000)	1.0000	
(12) SameIndustry	0.0010 (0.908)	-0.037* (0.000)	0.163* (0.000)	-0.0120 (0.155)	-0.069* (0.000)	0.216* (0.000)	0.163* (0.000)	-0.066* (0.000)	-0.181* (0.000)	0.423* (0.000)	-0.183* (0.000)	1.0000
(13) Hostility	-0.0020 (0.815)	0.031* (0.000)	0.018* (0.024)	-0.0120 (0.180)	-0.043* (0.000)	-0.023* (0.003)	-0.056* (0.000)	0.057* (0.000)	0.072* (0.000)	-0.0020 (0.760)	-0.0050 (0.530)	0.0020 (0.785)
(14) Management	-0.0110 (0.178)	-0.074* (0.000)	-0.033* (0.000)	-0.048* (0.000)	-0.042* (0.000)	-0.025* (0.002)	-0.054* (0.000)	0.178* (0.000)	0.125* (0.000)	-0.198* (0.000)	0.016* (0.045)	-0.165* (0.000)
(15) NumberofBidders	0.068* (0.000)	0.023* (0.003)	0.059* (0.000)	-0.0020 (0.803)	-0.023* (0.003)	0.057* (0.000)	0.0020 (0.842)	0.075* (0.000)	0.088* (0.000)	-0.020* (0.009)	-0.0100 (0.206)	-0.0050 (0.534)
(16) TotalAdvisors	-0.0160 (0.156)	0.129* (0.000)	0.506* (0.000)	0.092* (0.000)	0.037* (0.001)	0.033* (0.004)	0.117* (0.000)	0.130* (0.000)	0.064* (0.000)	-0.059* (0.000)	0.074* (0.000)	-0.035* (0.002)
(17) Toehold	-0.078* (0.000)	0.0090 (0.654)	0.046* (0.033)	0.0080 (0.711)	-0.045* (0.035)	-0.755* (0.000)	0.0290 (0.166)	-0.044* (0.039)	-0.048* (0.024)	0.0270 (0.207)	-0.0330 (0.122)	-0.0270 (0.198)

* $p < 0.1$, *P-values in parentheses*

Table 2 (*continued*)

Variables	(13)	(14)	(15)	(16)	(17)
(13) Hostility	1.0000				
(14) Management	-0.0100 (0.201)	1.0000			
(15) NumberofBidders	0.094* (0.000)	0.0140 (0.073)	1.0000		
(16) TotalAdvisors	0.024* (0.034)	-0.024* (0.035)	0.114* (0.000)	1.0000	
(17) Toehold	-0.091* (0.000)	-0.0110 (0.592)	-0.092* (0.000)	-0.0610 (0.060)	1.0000

* $p < 0.1$, *P-values in parentheses*

Table 3 Descriptive statistics for domestic versus cross border deals

	Domestic			Cross Border		
	N	Mean	SD	N	Mean	SD
CAR	12,060	0.0873	0.2719	4,163	0.1136	0.2663
CRET	12,060	0.1015	0.3041	4,163	0.1316	0.3053
BHAR	12,060	0.0732	0.2905	4,163	0.1064	0.2940
LogTotalAssets	11,275	5.5825	2.0552	3,935	5.5792	1.9845
BooktoMarket	10,081	2.3387	3.3234	3,592	2.6740	3.4855
HighTech	12,060	0.3066	0.4611	4,163	0.3651	0.4815
SharesAcquired	12,060	0.9124	0.1571	4,163	0.8875	0.1669
TargetStockOptions	12,060	0.3835	0.4863	4,163	0.2714	0.4448
BorrowingsFinancing	12,060	0.1094	0.3121	4,163	0.1415	0.3486
PercentageofCash	12,060	0.4679	0.4729	4,163	0.6924	0.4419
Acq Public	12,060	0.6074	0.4884	4,163	0.4398	0.4964
PEBidder	12,060	0.0463	0.2101	4,163	0.0680	0.2517
SameIndustry	12,060	0.5838	0.4929	4,163	0.5419	0.4983
Hostility	11,892	0.0194	0.1380	4,150	0.0299	0.1703
Management	12,060	0.0386	0.1927	4,163	0.0094	0.0963
NumberofBidders	12,060	1.0556	0.2724	4,163	1.0704	0.2918
TotalAdvisors	5,418	6.4719	2.7984	2,283	7.3092	3.2176
Toehold	1,628	0.2135	0.1490	601	0.2166	0.1385
Targ LogGDP	11,809	8.5090	1.2073	4,017	7.8268	1.4092
Acq LogGDP	11,791	8.5008	1.2197	3,941	7.2181	1.7398
GDPratio	11,791	0.9990	0.0270	3,941	0.9487	0.2803
Targ GDPperCapita	11,809	36.8296	16.5286	4,017	38.7070	18.4068
Acq GDPperCapita	11,791	36.8571	16.6133	3,941	39.9872	20.9581
GDPperCapitaRatio	11,791	1.0051	0.2885	3,941	1.8437	4.8723
Targ LogNetNI	11,795	8.3249	1.2342	4,010	7.6310	1.4464
Acq LogNetNI	11,773	8.3185	1.2418	3,836	7.1097	1.6315
NIratio	11,773	0.9992	0.0250	3,830	0.9627	0.2860
Targ NetNIperCapita	11,928	30.7401	13.6407	4,116	31.9108	14.6038
Acq NetNIperCapita	11,773	30.6438	13.6475	3,836	31.2303	15.1172
NIperCapitaRatio	11,769	1.0013	0.0931	3,829	1.7562	4.6047
ControlofCorruption	9,838	1.3258	0.5953	3,682	1.4217	0.6790
RegulatoryQuality	9,838	1.3488	0.4846	3,682	1.3969	0.5255
RuleofLaw	9,838	1.3715	0.4932	3,683	1.3973	0.5697
GovEffectiveness	9,838	1.4202	0.4475	3,682	1.4141	0.5174
PoliticalStability	9,838	0.5526	0.5393	3,683	0.5867	0.5513
MarketCaptoGDP	11,797	1.0874	0.5940	3,994	1.0635	0.7864
ListedCompanies	11,808	4.3347	2.3825	4,011	2.8539	2.3910

Table 4 Correlations of country-level variables

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
(1) CAR	1.0000													
(2) Targ_LogGDP	-0.024* (0.002)	1.0000												
(3) Acq_LogGDP	-0.034* (0.000)	0.708* (0.000)	1.0000											
(4) GDPPratio	-0.0100 (0.204)	-0.252* (0.000)	0.474* (0.000)	1.0000										
(5) Targ_GDPperCapita	-0.077* (0.000)	0.409* (0.000)	0.322* (0.000)	-0.038* (0.000)	1.0000									
(6) Acq_GDPperCapita	-0.085* (0.000)	0.315* (0.000)	0.231* (0.000)	-0.051* (0.000)	0.825* (0.000)	1.0000								
(7) GDPperCapitaRatio	0.017* (0.031)	-0.132* (0.000)	-0.101* (0.000)	0.039* (0.000)	-0.182* (0.000)	0.077* (0.000)	1.0000							
(8) Corruption	0.0040 (0.608)	0.152* (0.000)	0.141* (0.000)	0.0130 (0.139)	0.538* (0.000)	0.381* (0.000)	-0.261* (0.000)	1.0000						
(9) RegulatoryQuality	0.018* (0.040)	0.271* (0.000)	0.217* (0.000)	-0.029* (0.001)	0.595* (0.000)	0.433* (0.000)	-0.286* (0.000)	0.907* (0.000)	1.0000					
(10) RuleofLaw	-0.0120 (0.154)	0.314* (0.000)	0.255* (0.000)	-0.028* (0.001)	0.659* (0.000)	0.480* (0.000)	-0.277* (0.000)	0.931* (0.000)	0.916* (0.000)	1.0000				
(11) GovEffectiveness	0.0020 (0.773)	0.286* (0.000)	0.232* (0.000)	-0.028* (0.001)	0.554* (0.000)	0.378* (0.000)	-0.288* (0.000)	0.924* (0.000)	0.900* (0.000)	0.940* (0.000)	1.0000			
(12) PoliticalStability	0.021* (0.013)	-0.049* (0.000)	-0.0120 (0.150)	0.050* (0.000)	0.260* (0.000)	0.156* (0.000)	-0.230* (0.000)	0.703* (0.000)	0.634* (0.000)	0.686* (0.000)	0.690* (0.000)	1.0000		
(13) MarketCaptoGDP	0.025* (0.002)	0.190* (0.000)	0.151* (0.000)	-0.0130 (0.091)	0.284* (0.000)	0.213* (0.000)	-0.059* (0.000)	0.153* (0.000)	0.268* (0.000)	0.208* (0.000)	0.242* (0.000)	0.025* (0.004)	1.0000	
(14) ListedCompanies	0.054* (0.000)	0.699* (0.000)	0.489* (0.000)	-0.175* (0.000)	-0.065* (0.000)	-0.130* (0.000)	-0.066* (0.000)	0.150* (0.000)	0.266* (0.000)	0.241* (0.000)	0.281* (0.000)	0.056* (0.000)	0.124* (0.000)	1.0000

* $p < 0.1$, P -values in parentheses

Table 5 Descriptive statistics for alternative CAR windows

	N	Mean	Median	SD	Min	Max
CAR (-60, -1)	16,223	0.0941	0.0644	0.2707	-0.6859	1.1225
CAR (-40, -1)	16,223	0.0928	0.0645	0.2222	-0.5118	0.9437
CAR (-20, -1)	16,223	0.0709	0.0426	0.1697	-0.3579	0.7598

Table 5 shows summary statistics for the different abnormal return windows considered. Consistent with past literature and theoretical expectations, the run-up is smaller closer to the event date (i.e. for shorter windows). Furthermore, the range of values decreases for shorter windows, because for shorter periods of time more extreme returns are less likely.

Results and Discussions

Table 6 tests Hypothesis 1, namely testing whether target run-up is higher for cross border transactions than domestic deals, and attempting to show any differences in information leakage. The *Cross Border* dummy variable is statistically significant at the 1% level and positive across all model specifications, suggesting that pre-announcement CARs are on average higher for cross border deals. The size of this effect is also economically significant. Table 1 showed that the average CAR for the entire sample was 9.41%, whilst the models in Table 5 suggest the CAR is between 255 and 351 basis points higher when the deal is cross border. Model (1) presents only the dependent and independent variables of interest, and thus is subject to a high degree of omitted variable bias. However, when introducing deal level controls in specifications (2) and (3) the initial coefficient of 0.313 remains close to constant at 0.032 and 0.0266 respectively, showing that the cross-border nature of these deals still matters despite controlling for a variety of variables that have been shown to influence CAR, as discussed in previous sections.

Interestingly, controls that had strong theoretical justifications and have been used in previous literature such as *TargStockOptions*, *SameIndustry*, and *Acq_Public* are not statistically significant here. *Acq_Public*, was significantly positively correlated with CAR, as shown in Table 2, however the effect is likely through a larger size and thus captured within *LogTotalAssets*. For all firm- and deal- level controls, the sign of the coefficients are in keeping with the theoretical expectations outlined in the Methodology.

Specifications (4) and (5) include only target country-level controls. Variables with strong economic rationale such as *RuleofLaw*, *RegulatoryQuality*, and *MarketCaptoGDP* initially do not have a statistically significant effect on CAR. The level of corruption in the country does appear to have a statistically significant, positive effect on CAR, suggesting that for higher levels of corruption, CAR is 3.6 to 4.5 percentage points higher. *MarketCaptoGDP* which aims to measure the financial development of the target country is statistically insignificant across all models, whilst *ListedCompanies*, which can also measure financial development to a lesser extent, has a small, statistically significant positive effect on run-up.

Specifications (6), (7) and (8) make use of both deal- and country-level controls. Column (6) presents the most extensive range of control variables, although this is not the most preferred model due to some variables lacking statistical significance. Those control variables which were statistically significant in other model specifications are carried forward to specifications (7) and (8). (8) excludes those country-level controls for which there were fewer observations in order to avoid distortions from limiting the sample.

In relation to Hypothesis 1, these results show that cross border deals have an economically and statistically significantly higher level of pre-announcement target run-up, including when controlling for variables shown to determine run-up. From these models, inferences can also be made to information leakage. Management involvement, as shown by the variable *Management*, reduces CAR by approximately 3 percentage points, which represents one third of the mean CAR in this sample. This is in keeping with Brigida and Madura (2012) which hypothesised that management would avoid leaking information if they are involved in the deal as it comes at their own cost. Furthermore, PE bidders have approximately 2% lower CAR, arguably because of their superior transaction expertise and ability to minimise information leakage (Brigida & Madura, 2012). If the transaction is funded using borrowings, then CAR is around 2.5% higher, which could indicate that information is leaked when banks are involved. Variables related to deal anticipation such as the number of bidders, target size (*LogTotalAssets*), technology (*HighTech*) and target value (*BooktoMarket*) are also statistically significant in these models.

Table A3 in the appendix employs alternative independent variables (BHAR, CRET). The findings are in keeping with the results of Table 6. Table A3 also displays models without fixed effects, showing that the choice of statistical techniques did not distort the results. Finally, table A3 also

shows models that exclude target nations with a low number of observations, and also excluding targets in the USA as these firms make up approximately half of the sample (Table A1). The findings discussed above are robust to excluding target nations with less than 10 observations, and the USA.

Table 6 Regression of cross border dummy variable on target cumulative abnormal return (CAR)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cross Border	0.0313*** (0.0052)	0.032*** (0.0056)	0.0266*** (0.0057)	0.0351*** (0.0056)	0.0346*** (0.0056)	0.0285*** (0.0061)	0.0281*** (0.0061)	0.0255*** (0.0058)
LogTotalAssets		-0.0107*** (0.0013)	-0.0112*** (0.0014)			-0.0107*** (0.0015)	-0.0105*** (0.0014)	-0.0103*** (0.0014)
BooktoMarket		0.0042*** (0.0008)	0.0039*** (0.0008)			0.0044*** (0.0009)	0.0044*** (0.0008)	0.0041*** (0.0008)
TarStockOptions			0.0132 (0.0095)			0.0109 (0.0101)		
HighTech			0.0122** (0.0053)			0.0140** (0.0057)	0.0134** (0.0056)	0.0144*** (0.0054)
PEBidder			-0.0198** (0.0092)			-0.0175* (0.0093)	-0.0222** (0.0087)	-0.0239*** (0.0087)
Acq_Public			0.0070 (0.006)			0.0104 (0.0064)		
Borrowings			0.0239*** (0.006)			0.0292*** (0.0065)	0.0299*** (0.0065)	0.0248*** (0.0061)
PercentageofCash			0.0191*** (0.0061)			0.0177*** (0.0066)	0.0166*** (0.0062)	0.0183*** (0.0059)
NumberofBidders			0.0657*** (0.0098)			0.0733*** (0.0099)	0.0731*** (0.0099)	0.0652*** (0.0099)
Hostility			-0.0310** (0.0142)			-0.0255 (0.0173)		-0.0307** (0.0146)
SameIndustry			0.0000 (0.0054)			0.0001 (0.0058)		
SharesAcquired			0.0535*** (0.0191)			0.0447** (0.0206)	0.0596*** (0.0202)	0.0599*** (0.0197)
Management			-0.0251** (0.0122)			-0.0330** (0.0133)	-0.0379*** (0.0128)	-0.0323*** (0.0117)
RegulatoryQuality				0.0310 (0.0199)	0.0225 (0.0216)	0.0159 (0.0232)		
Corruption				0.0363* (0.0196)	0.0453** (0.0229)	0.0310 (0.0248)	0.0203 (0.0196)	
MarketCaptoGDP				0.0033 (0.0123)	0.0086 (0.0127)	0.0073 (0.0149)		0.0059 (0.0135)
RuleofLaw					-0.0179 (0.0353)	0.0120 (0.0404)		
GovEffectiveness					-0.0423 (0.0269)	-0.0424 (0.0303)		
PoliticalStability					0.0480*** (0.0146)	0.0419*** (0.0158)	0.0364** (0.0144)	
ListedCompanies					0.0080** (0.0036)	0.0049 (0.0043)	0.0054 (0.0042)	0.0082** (0.0037)
_cons	0.3376*** (0.1099)	0.1546*** (0.0202)	0.0209 (0.0303)	0.0050 (0.0336)	-0.0115 (0.0429)	-0.0815 (0.0538)	-0.0818* (0.0432)	-0.0332 (0.0364)
Observations	16,223	13,639	13,507	13,494	13,494	11,795	11,917	13,163
R-squared	0.0422	0.0588	0.0685	0.0449	0.0462	0.0727	0.0716	0.0681
Target Nation FEs	YES	YES	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Table 7 presents the same regression model as in column (8) from Table 6, but also the same regression model using alternative CAR windows. In general, the variable coefficients are lower for shorter CAR windows, in keeping with the lower averages as shown in Table 5. However, *MarketCaptoGDP* and *ListedCompanies* become higher for shorter windows, possibly implying that more developed financial markets price in the upcoming deal closer to the announcement.

Table 7 Regression of cross border dummy variable on alternative CAR windows

	(1) CAR (-20, -1)	(2) CAR (-40, -1)	(3) CAR (-60, -1)
Cross Border	0.0174*** (0.0046)	0.0229*** (0.0059)	0.0255*** (0.0058)
LogTotalAssets	-0.0060*** (0.0011)	-0.0088*** (0.0014)	-0.0103*** (0.0014)
BooktoMarket	-0.0100* (0.0006)	0.0010 (0.0008)	0.0041*** (0.0008)
HighTech	0.0216*** (0.0040)	0.0177*** (0.0053)	0.0144*** (0.0054)
PEBidder	-0.0170** (0.0072)	-0.0121 (0.0093)	-0.0239*** (0.0087)
Borrowings	0.0067 (0.0047)	0.0143** (0.0059)	0.0248*** (0.0061)
PercentageofCash	0.0084* (0.0044)	0.0116** (0.0058)	0.0183*** (0.0059)
NumberofBidders	0.0238*** (0.0069)	0.0417*** (0.0083)	0.0652*** (.00099)
Hostility	0.0010 (0.0116)	-0.0155 (0.0152)	-0.0307** (0.0146)
SharesAcquired	0.0297 (0.0184)	0.0603** (0.0237)	0.0599*** (0.0197)
Management	-0.0414*** (0.0084)	-0.0444*** (0.0112)	-0.0323*** (0.0117)
MarketCaptoGDP	0.0256** (0.0106)	0.0352* (0.0201)	0.0059 (0.0135)
ListedCompanies	0.0114*** (0.0036)	0.0099** (0.0047)	0.0082** (0.0037)
_cons	-0.0164 (0.0335)	-0.0436 (0.0421)	-0.0332 (0.0364)
Observations	13,163	13,163	13,163
R-squared	0.0597	0.0657	0.0681
Target Nation FEs	YES	YES	YES
Year FEs	YES	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Table 8 Regression of target GDP on CAR for cross-border deals

	(1)	(2)	(3)	(4)
Targ_LogGDP	0.0866*** (0.0289)	0.0657** (0.0328)	0.0590* (0.0344)	0.0736** (0.0351)
RegulatoryQuality		0.0304 (0.0364)	-0.0064 (0.0414)	
Corruption		0.0923** (0.0374)	0.0673* (0.0398)	0.0776** (0.0384)
MarketCaptoGDP		0.0005 (0.0177)	-0.0022 (0.0187)	0.0008 (0.0192)
RuleofLaw			0.0563 (0.0655)	
GovEffectiveness			0.0119 (0.0504)	
PoliticalStability			0.0273 (0.0251)	0.0420* (0.0249)
ListedCompanies			0.0123 (0.0082)	
LogTotalAssets				-0.0110*** (0.0027)
BooktoMarket				0.0057*** (0.0015)
HighTech				0.0207** (0.0104)
PEBidder				-0.0237* (0.0143)
Borrowings				0.0423*** (0.0118)
PercentageofCash				0.0361*** (0.0127)
NumberofBidders				0.1091*** (0.0181)
Hostility				-0.0567** (0.0238)
SharesAcquired				0.1086*** (0.0380)
Management				-0.0207 (0.0481)
_cons	-0.6699*** (0.1807)	-0.5343** (0.2241)	-0.5631** (0.2490)	-0.7422*** (0.2471)
Observations	4,017	3,667	3,667	3,221
R-squared	0.0668	0.0657	0.0674	0.1186
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

*Robust standard errors are in parentheses**** $p < .01$, ** $p < .05$, * $p < .1$ **Table 9** Regression of target GDP per capita on CAR for cross-border deals

	(1)	(2)	(3)	(4)
Targ_GDPperCapita	0.0011 (0.0008)	0.0010 (0.0009)	0.0014 (0.0009)	0.0020** (0.0009)
RegulatoryQuality		0.0426 (0.0354)	-0.0066 (0.0413)	
Corruption		0.1016*** (0.0370)	0.0733* (0.0395)	0.0956** (0.0373)
MarketCaptoGDP		0.0004 (0.0177)	-0.0028 (0.0188)	-0.0003 (0.0193)
RuleofLaw			0.0382 (0.0634)	
GovEffectiveness			0.0281 (0.0501)	
PoliticalStability			0.0386 (0.0250)	0.0565** (0.0244)
ListedCompanies			0.0143* (0.0082)	
LogTotalAssets				-0.0113*** (0.0027)
BooktoMarket				0.0055*** (0.0015)
HighTech				0.0202* (0.0104)
PEBidder				-0.0241* (0.0143)
Borrowings				0.0422*** (0.0118)
PercentageofCash				0.0360*** (0.0127)
NumberofBidders				0.1094*** (0.0181)
Hostility				-0.0561** (0.0238)
SharesAcquired				0.1135*** (0.0381)
Management				-0.0202 (0.0480)
_cons	-0.1334*** (0.0069)	-0.1152 (0.0713)	-0.1964** (0.0939)	-0.2979*** (0.0913)
Observations	4,017	3,667	3,667	3,221
R-squared	0.0649	0.0650	0.0673	0.1186
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

*Robust standard errors are in parentheses**** $p < .01$, ** $p < .05$, * $p < .1$

Tables 8 and 9 test Hypothesis 2.1 by regressing a development measure together with controls on the pre-announcement target CAR for cross-border deals only. Table 8 uses the *Targ_LogGDP* as the independent variable of interest, whereas Table 9 uses *Targ_GDPperCapita*. Both use a combination of country-level control variables in columns (2) and (3), as well as the full model used in Hypothesis 1 that was found to most accurately model CAR.

The overall size of the target's economy, as measured by *Targ_LogGDP*, appears to be positively related with the target pre-announcement CAR. This is statistically significant across all model specifications, including when controlling for country- and deal-level variables. A one standard deviation increase would result in a 10% higher CAR according to the first model specification. This is lower when accounting for the effect of Corruption, as well as other country- and deal-level controls, but always representing at least a 7.1% increase. This goes against theoretical intuition, which suggested that more developed economies would have better institutional and regulatory structures, limiting information leakage. One possible explanation for this positive relationship is that companies in less developed countries are more difficult to trade or find information on, and thus, even with the will to make informed trades, it could be harder to actually make this, especially on a scale that would influence the stock's CAR. Another possibility, as discussed in the Methodology, is that the true relationship is non-linear. This is more with Figure 1 and Table 10.

With regards to the target GDP per capita, the coefficient seems both statistically insignificant and not strongly economically significant, with a one standard deviation change in GDP per capita representing around a 1.7% increase in CAR, depending on the controls included. However, the coefficient gains statistical significance at the 5% level when controlling for key deal-level variables, as shown in specification (4). It also gains in economic significance, with a standard deviation change representing a 3.4% increase in CAR. Again, this positive relationship goes against expectations, but could be justified with similar arguments as to those for the economy as a whole.

Figure 1 and Table 10 attempt to explain the counterintuitive finding with respect to the relationship between target GDP and target run-up. Graphing target GDP and log GDP against the pre-announcement run-up using a polynomial function appears to show a parabola with a negative slope. This is consistent with the argumentation presented in the methodology. For countries with low levels of development, access to trading may be limited, thus reducing the ability for informed

individuals to profit from leaked information. As development increases, so too would access to capital markets, and so too would information leakage, up until a point where institutional and regulatory development catch up to reduce the opportunity to leak information and make inside trades. Figure 1 is split into four panels: the left using the full sample, whilst the right only uses cross-border deals. Furthermore, the top two panels display GDP whilst the bottom two show the log transformed GDP. The top left graph appears to be the most reliable given that the confidence interval is the narrowest. With the exception of the bottom right panel, each graph is consistent with the argumentation presented, whilst the bottom right panel displays a relationship that is more similar to the results in Table 8.

Table 10 attempts to model this relationship using non-linear terms in the regression framework used previously, namely the squared and cubed target GDP.

Figure 1 Polynomial plots of Target GDP and log GDP against CAR with 95% confidence intervals

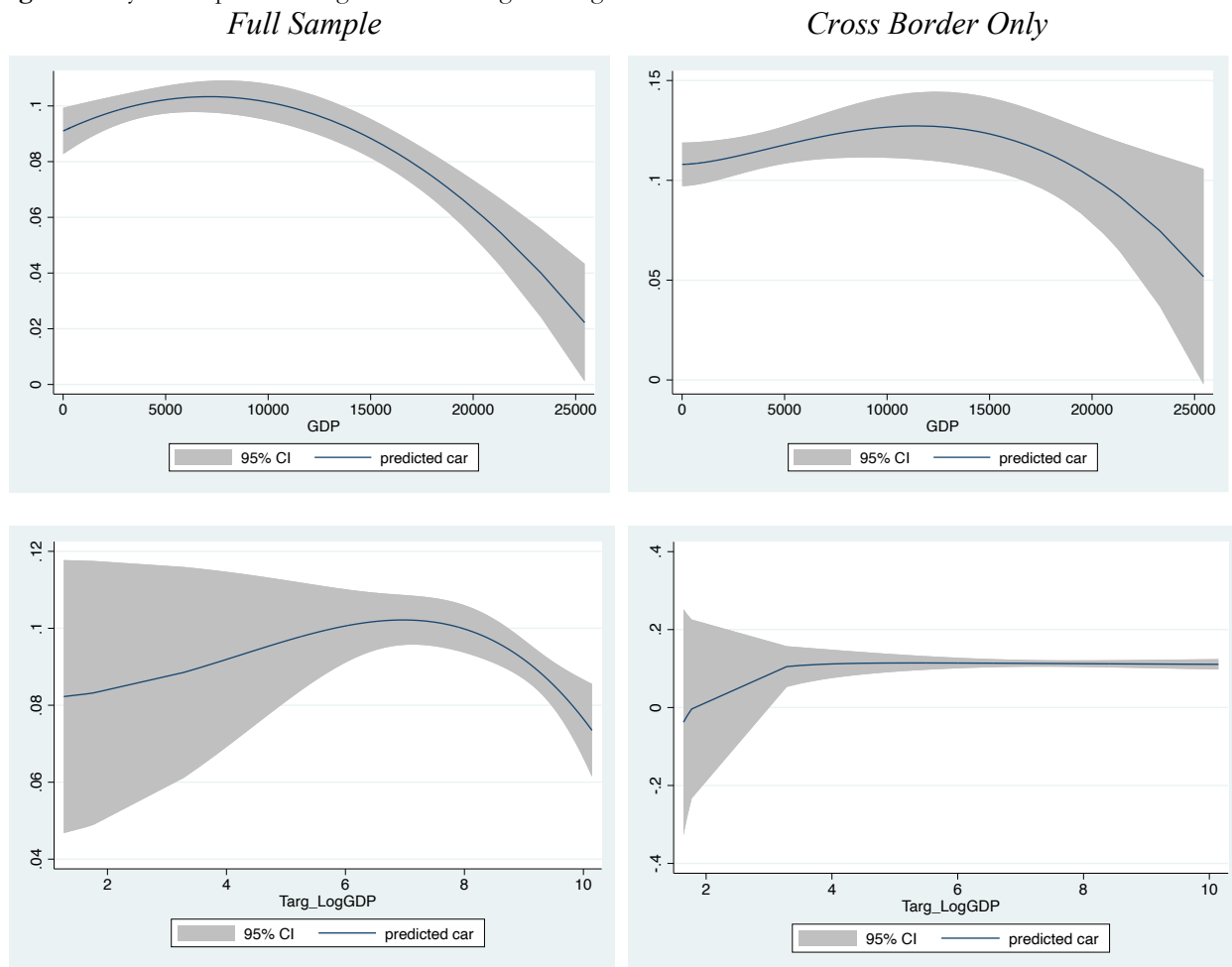


Table 10 Non-linear regression of target GDP on CAR

	(1)	(2)	(3)	(4)	(5)	(6)
Targ_GDPt	-0.0027** (0.0012)	-0.0102*** (0.0029)	-0.0043*** (0.0014)	-0.0100*** (0.0032)	-0.0013 (0.0056)	-0.0012 (0.0065)
Targ_GDPt_Squared	0.0001 (0.0001)	0.0010*** (0.0003)	0.0001* (0.0001)	0.0008** (0.0003)	0.0001 (0.0006)	-0.0001 (0.0007)
Targ_GDPt_Cubed		-0.0000*** (0.0000)		-0.0000* (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)
LogTotalAssets			-0.0121*** (0.0015)	-0.0121*** (0.0015)		-0.0128*** (0.0027)
BooktoMarket			0.0045*** (0.0008)	0.0045*** (0.0008)		0.0054*** (0.0015)
HighTech			0.0134** (0.0056)	0.0136** (0.0056)		0.0159 (0.0103)
PEBidder			-0.0241*** (0.0087)	-0.0239*** (0.0087)		-0.0234 (0.0144)
BorrowingsFinan~g			0.0323*** (0.0064)	0.0319*** (0.0064)		0.0411*** (0.0115)
PercentageofCash			0.0199*** (0.0060)	0.0193*** (0.0060)		0.0292** (0.0127)
NumberofBidders			0.0735*** (0.0098)	0.0731*** (0.0098)		0.1104*** (0.0172)
SharesAcquired			0.0617*** (0.0185)	0.0571*** (0.0188)		0.0977*** (0.0348)
Management			-0.0462*** (0.0126)	-0.0438*** (0.0126)		-0.0055 (0.0454)
ControlofCorrup~n			-0.0242*** (0.0069)	-0.0231*** (0.0070)		-0.0408*** (0.0126)
PoliticalStabil~y			-0.0069 (0.0084)	-0.0071 (0.0084)		0.0100 (0.0163)
MarketCaptoGDP			0.0180*** (0.0063)	0.0174*** (0.0063)		0.0203*** (0.0071)
_cons	0.1047*** (0.0047)	0.1154*** (0.0060)	0.0342 (0.0216)	0.0461** (0.0228)	0.1141*** (0.0094)	-0.0276 (0.0392)
Observations	15,826	15,826	11,909	11,909	4,017	3,230
R-squared	0.0271	0.0277	0.0580	0.0583	0.0407	0.0902
Target Nation FEs	NO	NO	NO	NO	NO	NO
Year FEs	YES	YES	YES	YES	YES	YES
Cross Border Only	NO	NO	NO	NO	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Note. GDPt refers to GDP measured in trillions, rather than billions as used in the rest of this study. This is in order to scale GDP to have meaningful coefficients, which was previously not an issue when using the natural logarithm. Target Nation fixed effects are removed in order to maximise the variation in countries to be analysed.

Table 10 uses two model setups, namely a quadratic function and a cubic function. These are improved by using deal- and country- control variables. Columns (5) and (6) make use of only the cross-border portion of the sample, whereas the others use the full sample. None of the GDP terms are statistically significant when looking only at the cross-border sample, which may be a result of the limited sample size, or simply that there is not a relationship for cross-border deals. However, when looking at the sample as a whole, it appears that the data fits a cubic function. From Figure 1 and the theoretical discussion, we would expect to find a negative coefficient for the squared term

in a quadratic equation, or a negative term for at least one of the squared or cubed term in a cubic function. This is not true for the quadratic functions displayed in Table 10, although target GDP does now show a negative coefficient. The cubic function, as shown with control variables in column (4) has a cubed term with a negative coefficient, although this coefficient is small enough that, rounded to four decimal points, it is equal to zero (one would have to round to 6 decimal points before getting a non-zero value). The low significance of this model is most likely due to its setup. Further investigation would require more complex modelling techniques that go beyond the scope of this study. What is shown however is that there is room for a deeper look. The relationship between development is not as simple as first posited in the Theoretical Framework, and Table 10 and Figure 1 provide an indication that further investigation into non-linear relationships is required.

Table 11 and 12 test Hypothesis 2.2, namely investigating the relationship between acquiror development and run-up. Using the same setup as for testing Hypothesis 2.1, Table 11 displays analysis using GDP whilst Table 12 uses GDP per capita. The coefficient for the acquiring country's GDP is slightly negative and statistically insignificant across all specifications. The negative coefficient is contrary to the expectation that higher acquiror GDP will have higher run-up, however the magnitude and statistical insignificance of these coefficients suggest instead that there is no relationship. Furthermore, when controlling for all considered target country variables, the acquiror country GDP variable has a positive coefficient. The conclusion derived from these tests therefore that there is no significant relationship between acquiror GDP and target run-up.

Table 12 shows that, across all model specifications, the acquiror GDP per capita has a statistically significant, negative relationship with run-up. Whilst acquiror development as a whole was expected to have a positive relationship with run-up, this suggests that at a per capita level the opposite is true. This could be justified using the incentive motivation from the Theoretical Framework, namely that if people have lower wealth to start with then they are more driven to make the additional profits that informed trading could afford them. This argument could be true for both acquiror and target nation, given that both have advance knowledge of the impending event. The economic significance of this finding however is very low, with a standard deviation increase in GDP per capita resulting in only a 0.89% change in CAR.

Table 11 Regression of acquiror GDP on CAR for cross-border deals

	(1)	(2)	(3)	(4)
Acq_LogGDP	-0.0005 (0.0025)	-0.0002 (0.0026)	0.0002 (0.0026)	-0.0028 (0.0028)
RegulatoryQuality		0.0542 (0.0345)	0.0116 (0.0394)	
Corruption		0.0964*** (0.0372)	0.0680* (0.0398)	0.0887** (0.0374)
MarketCapToGDP		-0.0045 (0.0195)	-0.0075 (0.0203)	-0.0046 (0.0212)
RuleofLaw			0.0306 (0.0630)	
GovEffectiveness			0.0240 (0.0504)	
PoliticalStability			0.0325 (0.0247)	0.0476** (0.0239)
ListedCompanies			0.0154* (0.0082)	
LogTotalAssets				-0.0111*** (0.0027)
BooktoMarket				0.0056*** (0.0015)
HighTech				0.0196* (0.0106)
PEBidder				-0.0245 (0.0149)
Borrowings				0.0418*** (0.0120)
PercentageofCash				0.0348*** (0.0128)
NumberofBidders				0.1066*** (0.0182)
Hostility				-0.0539** (0.0239)
SharesAcquired				0.1178*** (0.0382)
Management				-0.0218 (0.0500)
_cons	-0.1276*** (0.0147)	-0.0955 (0.0725)	-0.1614* (0.0927)	-0.2147** (0.0917)
Observations	3,941	3,596	3,596	3,161
R-squared	0.0634	0.0636	0.0658	0.1171
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

*Robust standard errors are in parentheses**** $p < .01$, ** $p < .05$, * $p < .1$ **Table 12** Regression of acquiror GDP per capita on CAR for cross-border deals

	(1)	(2)	(3)	(4)
Acq_GDPperCapita	-0.0005** (0.0002)	-0.0005** (0.0002)	-0.0005** (0.0002)	-0.0006** (0.0003)
RegulatoryQuality		0.0584* (0.0346)	0.0170 (0.0398)	
Corruption		0.0935** (0.0373)	0.0661* (0.0398)	0.0876** (0.0375)
MarketCapToGDP		-0.0051 (0.0195)	-0.0080 (0.0203)	-0.0049 (0.0211)
RuleofLaw			0.0277 (0.0632)	
GovEffectiveness			0.0240 (0.0504)	
PoliticalStability			0.0316 (0.0247)	0.0462* (0.0239)
ListedCompanies			0.0149* (0.0083)	
LogTotalAssets				-0.0112*** (0.0027)
BooktoMarket				0.0057*** (0.0015)
HighTech				0.0194* (0.0105)
PEBidder				-0.0218 (0.0149)
Borrowings				0.0423*** (0.0120)
PercentageofCash				0.0344*** (0.0128)
NumberofBidders				0.1072*** (0.0183)
Hostility				-0.0556** (0.0240)
SharesAcquired				0.1138*** (0.0381)
Management				-0.0252 (0.0496)
_cons	-0.1250*** (0.0066)	-0.0848 (0.0709)	-0.1457 (0.0917)	-0.2106** (0.0912)
Observations	3,941	3,596	3,596	3,161
R-squared	0.0645	0.0648	0.0668	0.1181
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

*Robust standard errors are in parentheses**** $p < .01$, ** $p < .05$, * $p < .1$

Table 13 Regression of ratio of GDP on CAR

	(1)	(2)	(3)	(4)
Int_GDPratioxCB	0.0037*** (0.0007)	0.0039*** (0.0007)	0.0030*** (0.0007)	-0.0009 (0.0015)
RegulatoryQuality		0.0320 (0.0199)		
Corruption		0.0358* (0.0197)	0.0302 (0.0184)	0.0303* (0.0184)
MarketCaptoGDP		0.0009 (0.0127)	0.0026 (0.0150)	0.0027 (0.0151)
LogTotalAssets			-0.0105*** (0.0015)	-0.0104*** (0.0015)
BooktoMarket			0.0046*** (0.0009)	0.0046*** (0.0009)
HighTech			0.0146** (0.0057)	0.0148*** (0.0057)
PEBidder			-0.0230*** (0.0088)	-0.0220** (0.0088)
Borrowings			0.0310*** (0.0065)	0.0303*** (0.0065)
PercentageofCash			0.0156** (0.0062)	0.0145** (0.0062)
NumberofBidders			0.0725*** (0.0099)	0.0726*** (0.0099)
Hostility			-0.0232 (0.0174)	-0.0240 (0.0174)
SharesAcquired			0.0533*** (0.0206)	0.0553*** (0.0206)
Management			-0.0402*** (0.0128)	-0.0382*** (0.0128)
PoliticalStability			0.0374*** (0.0145)	0.0375*** (0.0145)
Cross Border				0.0349*** (0.0127)
_cons	0.3363*** (0.1099)	0.0077 (0.0337)	-0.0586 (0.0437)	-0.0621 (0.0438)
Observations	15,732	13,409	11,725	11,725
R-squared	0.0423	0.0445	0.0717	0.0723
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Note. Int_GDPratioxCB is the ratio of log GDPs, multiplied by the acquiror log GDP, interacted with the cross border dummy variable.

Table 14 Regression of ratio of GDP per capita on CAR

	(1)	(2)	(3)	(4)
Int_GDPperCapitaRatioxCB	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
RegulatoryQuality		0.0304 (0.0200)		
Corruption		0.0359* (0.0197)	0.0302 (0.0184)	0.0303* (0.0184)
MarketCaptoGDP		-0.0001 (0.0130)	0.0018 (0.0152)	0.0027 (0.0150)
LogTotalAssets			-0.0101*** (0.0015)	-0.0105*** (0.0015)
BooktoMarket			0.0047*** (0.0009)	0.0046*** (0.0009)
HighTech			0.0163*** (0.0057)	0.0146*** (0.0057)
PEBidder			-0.0230*** (0.0088)	-0.0221** (0.0088)
Borrowings			0.0308*** (0.0065)	0.0304*** (0.0065)
PercentageofCash			0.0196*** (0.0061)	0.0144** (0.0062)
NumberofBidders			0.0721*** (0.0100)	0.0726*** (0.0099)
Hostility			-0.0233 (0.0174)	-0.0238 (0.0174)
SharesAcquired			0.0551*** (0.0207)	0.0546*** (0.0207)
Management			-0.0439*** (0.0128)	-0.0382*** (0.0128)
PoliticalStability			0.0375*** (0.0145)	0.0375*** (0.0145)
Cross Border				0.0294*** (0.0062)
_cons	0.3388*** (0.1099)	0.0153 (0.0337)	-0.0595 (0.0439)	-0.0612 (0.0438)
Observations	15,732	13,409	11,725	11,725
R-squared	0.0406	0.0424	0.0705	0.0723
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Note. Int_GDPperCapitaRatioxCB is the ratio of GDP per capita, multiplied by the acquiror GDP per capita, interacted with the cross border dummy variable.

Finally, Tables 13 and 14 show the results from the models testing Hypothesis 2.3. Here, the independent variables of interest are respectively the ratio of log GDP and the ratio of GDP per capita, both multiplied by the corresponding measure from the acquiror's nation. This is then interacted with the cross-border dummy variable to show the relationship given that the deal is cross-border.

Table 13 initially appears to show that the relative size of the economies does have an effect, specifically that when the acquiror nation is larger than the target in a cross-border transaction the run-up is higher. This is in keeping with the hypothesis that for a larger difference in development there is a greater amount of information leakage. However, model specification (4) shows that, when the cross-border dummy variable is included in addition to the measure of relative development, the statistical and economic significance of this measure is removed. Thus, Table 13 suggests that the relative development of the economies as a whole does not significantly influence run-up. Table 14 reinforces the rejection of Hypothesis 2.3 by showing that, at a per capita level, there is also no significant relationship to be derived.

Conclusion

This study investigates target run-up and information leakage in cross-border M&A transactions. A review of literature in the field and research on related topics showed that, whilst many studies investigate international M&A activity or information leakage and run-up, there are few that explicitly explore information leakage in a cross-border context. This called for further investigation due to its significance for regulators, governments, and company management.

A conceptual framework is derived with which to look at the causes of information leakage, namely opportunity and incentive. This is used to develop an empirical framework with which to look at run-up and information leakage in domestic and cross-border deals. Using multivariate regression models, the relationship between deal-level and country-level variables were assessed to explain the determinants of run-up, whilst also including the variables of interest to assess their impact.

The first research question of this study asked whether information leakage is more prevalent from cross-border M&A transactions. The results showed conclusively that, for cross-border transactions, target stock price run-up was higher by at least 2.5%, controlling for deal- and country-level characteristics. Using a broad interpretation of information leakage, in that it

constitutes any way in which outsiders know a deal will happen before it happens, this can be taken as evidence for higher information leakage in cross-border transactions. Control variables such as management involvement and borrowings financing also provide evidence in favour of the information leakage framework with which to assess and interpret run-up. The finding of higher cross-border run-up was also robust to including key target nation controls, such as financial development, corruption and political stability.

This result holds significance for real world practitioners. Firstly, for regulators aiming to limit insider trading, increased attention to trades surrounding cross-border M&A deals may reduce the extent of information leakage. Secondly, managers considering acquisition targets should pay additional attention to information security when considering a cross-border transaction, and, given that run-up acts as a cost to acquiror, possibly favour domestic deals to limit their costs.

The second research question asked whether the development of the target and acquiror's nations affected the degree of information leakage. Linear regression models showed a positive impact on information leakage if the target country has a higher GDP. Other measures of development were shown to be either economically or statistically insignificant. Further investigation into the role of the target country's development suggested that there may be a non-linear relationship between the size of the economy and the level of information leakage. Namely, for small, less developed economies information leakage is low, possibly due to limited access to stock markets, but then information leakage is increasing in development. For more developed economies, information leakage starts to be decreasing in development, possibly due to more developed regulation limiting the scope for informed trading. The statistical analysis on this relationship was however inconclusive.

Further study on cross-border M&A and information leakage could look at regional and cultural breakdowns of this result in assessing what causes information leakage to be higher for cross-border deals. With respect to the development angle, implementing alternative, non-linear models could result in better insights into the role development plays. Lastly, the relationship with premia and run-up for cross-border deals, as well as post-merger performance should be assessed to build a more complete picture of this field.

Appendices

Table A1 Cumulative Abnormal Return (CAR) and GDP by Target Nation

Target Nation	N	Cross Border	Mean CAR	Median CAR	St. Dev. CAR	Mean GDP (bn)
Australia	863	41%	0.0911	0.0583	0.2723	1,146.902
Austria	20	80%	0.1107	0.0732	0.2778	385.3796
Belgium	49	65%	0.1124	0.0386	0.3049	456.0312
Brazil	64	41%	0.0469	0.0407	0.2513	1,688.413
Canada	65	71%	0.0793	0.0851	0.2592	1,220.633
Chile	37	49%	-0.0280	0.0249	0.2065	230.999
China (Mainland)	110	19%	0.0051	-0.0067	0.2269	7,499.933
Colombia	18	44%	0.0197	0.0531	0.3271	268.921
Denmark	94	39%	0.0602	0.0166	0.2946	283.2125
Egypt	2	0%	0.2155	0.2155	0.6373	305.5954
Estonia	1	100%	-0.0920	-0.0920	.	26.6341
Finland	60	65%	0.0254	0.0270	0.1719	234.0881
France	383	41%	0.0504	0.0455	0.2249	2,386.371
Germany	336	57%	0.1050	0.0568	0.2908	3,191.729
Greece	55	36%	0.0319	0.0329	0.2133	237.3933
Guernsey	4	100%	0.0877	0.0457	0.1649	(missing)
Hong Kong	54	48%	0.2683	0.1894	0.3734	219.5883
Hungary	11	64%	0.1638	0.1122	0.2472	116.8731
India	316	19%	0.0689	0.0310	0.2949	2,181.168
Indonesia	95	63%	0.1649	0.0672	0.3509	801.8289
Ireland	33	61%	0.1764	0.1681	0.3655	266.8573
Isle of Man	2	100%	-0.0946	-0.0946	0.1803	5.4869
Italy	159	40%	0.0668	0.0409	0.2046	1,931.805
Japan	1,203	9%	0.0360	0.0184	0.2421	4,967.911
Jersey	3	100%	0.2523	0.1048	0.3522	(missing)
Luxembourg	1	0%	0.2411	0.2411	.	60.0716
Malaysia	186	25%	0.1569	0.0962	0.2461	239.6456
Mexico	22	32%	0.0614	0.0470	0.1302	1,040.972
Monaco	1	100%	-0.0629	-0.0629	.	5.8758
Myanmar	1	100%	1.1225	1.1225	.	62.2635
Netherlands	115	55%	0.0650	0.0353	0.2634	714.8994
New Zealand	67	63%	0.0411	0.0556	0.2551	139.2002
Norway	128	52%	0.0452	0.0496	0.3072	403.787
Papua New Guinea	2	50%	0.0669	0.0669	0.0022	4.3509
Peru	1	100%	0.0430	0.0430	.	201.1755
Philippines	56	27%	0.2440	0.2041	0.3451	223.6496
Poland	214	45%	0.1300	0.0571	0.3437	501.5657
Portugal	15	47%	0.1715	0.0756	0.3533	183.8312
Russia	2	100%	0.0474	0.0474	0.3401	(missing)
Singapore	243	44%	0.1560	0.1041	0.2696	(missing)
South Africa	102	35%	0.0661	0.0317	0.2262	352.8528
South Korea	376	9%	0.0910	0.0439	0.3521	1,362.249
Spain	84	36%	0.0552	0.0585	0.1980	1,163.726
Sweden	248	49%	0.0475	0.0257	0.2335	450.9629
Switzerland	88	53%	0.0948	0.0641	0.2066	536.9824
Taiwan	145	21%	0.0871	0.0640	0.1898	(missing)
Thailand	93	29%	0.2258	0.1676	0.3203	311.8076
Turkey	89	34%	0.1418	0.1008	0.2852	815.6846
United Kingdom	1,306	50%	0.1070	0.0837	0.2701	2,355.335
United States	8,601	16%	0.1014	0.0754	0.2681	1,1885.74

Note. Some nations are not included within the World Bank Database, hence missing country-level variables

Table A2. Summary statistics: CAR by Year

Year	N	Cross Border	Mean CAR	Median CAR	CAR St Dev
1978	3	0.0%	33.8%	23.7%	23.2%
1979	2	0.0%	-2.2%	-2.2%	18.0%
1980	8	12.5%	12.3%	19.6%	15.1%
1981	65	15.4%	15.0%	14.5%	25.3%
1982	81	8.6%	7.9%	6.0%	29.3%
1983	107	10.3%	13.3%	12.7%	22.2%
1984	132	7.6%	7.7%	7.9%	22.4%
1985	123	5.7%	11.9%	10.1%	20.5%
1986	171	17.5%	9.8%	6.8%	20.6%
1987	167	16.8%	11.7%	10.2%	24.9%
1988	222	19.4%	17.3%	13.1%	24.3%
1989	201	25.4%	8.0%	5.0%	29.0%
1990	121	19.0%	7.2%	4.2%	32.5%
1991	129	7.0%	11.0%	8.1%	41.3%
1992	122	5.7%	14.1%	10.7%	34.4%
1993	159	14.5%	20.2%	11.9%	42.5%
1994	217	12.4%	14.1%	10.3%	26.6%
1995	289	17.6%	10.3%	7.1%	29.5%
1996	326	13.8%	11.3%	8.2%	24.5%
1997	457	15.5%	11.8%	9.8%	23.7%
1998	575	19.0%	7.1%	7.4%	33.8%
1999	718	26.7%	20.1%	15.3%	34.7%
2000	698	28.9%	16.5%	13.5%	36.6%
2001	559	21.3%	17.4%	15.6%	32.6%
2002	419	20.8%	14.6%	12.2%	43.0%
2003	489	21.7%	12.5%	8.3%	33.8%
2004	416	22.1%	5.8%	4.5%	29.6%
2005	546	28.0%	6.6%	5.3%	25.6%
2006	623	25.7%	4.3%	4.1%	20.3%
2007	661	35.6%	4.8%	4.3%	24.2%
2008	437	33.0%	6.1%	4.9%	43.7%
2009	463	25.3%	11.7%	7.8%	43.3%
2010	543	31.9%	8.4%	5.2%	30.8%
2011	552	28.4%	7.6%	4.7%	37.7%
2012	481	29.9%	4.2%	3.7%	27.8%
2013	430	27.9%	6.0%	2.6%	31.6%
2014	456	32.2%	6.0%	3.1%	25.6%
2015	523	31.9%	7.6%	3.7%	33.2%
2016	486	33.3%	10.4%	6.2%	38.9%
2017	453	31.8%	7.4%	3.2%	32.9%
2018	480	30.0%	4.8%	3.5%	26.5%
2019	465	34.0%	6.1%	2.9%	39.5%
2020	384	28.9%	11.7%	4.2%	39.9%
2021	482	28.0%	8.7%	5.5%	25.3%
2022	402	26.9%	8.0%	5.3%	34.1%
2023	376	31.6%	3.6%	2.7%	26.5%
2024	12	41.7%	10.0%	-4.0%	35.3%

Table A3 Hypothesis 1 robustness tests

<i>Dependent variable</i>	(1) BHAR	(2) BHAR	(3) CRET	(4) CRET	(5) CAR	(6) CAR	(7) CAR	(8) † CAR	(9) †† CAR
Cross Border	0.0349*** (0.0057)	0.0289*** (0.0066)	0.0340*** (0.0058)	0.0281*** (0.0068)	0.0301*** (0.0057)	0.035*** (0.0057)	0.0284*** (0.0061)	0.0285*** (0.0061)	0.0257*** (0.0081)
LogTotalAssets		-0.0039** (0.0015)		-0.0038** (0.0016)	-0.0144*** (0.0014)	-0.0131*** (0.0014)	-0.0108*** (0.0015)	-0.0104*** (0.0015)	-0.0125*** (0.0019)
BooktoMarket		0.0066*** (0.0010)		0.0072*** (0.0010)	0.0043*** (0.0009)	0.0045*** (0.0009)	0.0042*** (0.0009)	0.0045*** (0.0009)	0.0024** (0.0012)
HighTech		0.0093 (0.0061)		0.0082 (0.0063)	0.0083 (0.0057)	0.0097* (0.0056)	0.0153*** (0.0057)	0.0150*** (0.0057)	0.0148* (0.0085)
PEBidder		-0.0284*** (0.0088)		-0.0269*** (0.0091)	-0.0419*** (0.0087)	-0.0249*** (0.0087)	-0.0279*** (0.0087)	-0.0222** (0.0087)	-0.0148 (0.0117)
Borrowings		0.0354*** (0.0073)		0.0334*** (0.0076)	0.0314*** (0.0065)	0.0326*** (0.0064)	0.0257*** (0.0066)	0.0297*** (0.0065)	0.0403*** (0.0098)
PercentageofCash		0.0265*** (0.0066)		0.0267*** (0.0068)	-0.0002 (0.0061)	0.0109* (0.0062)	0.0078 (0.0062)	0.0150** (0.0062)	0.0126 (0.0085)
NumberofBidders		0.0824*** (0.0116)		0.0849*** (0.0122)	0.0765*** (0.0096)	0.0759*** (0.0098)	0.0753*** (0.0098)	0.0731*** (0.0099)	0.0953*** (0.0137)
Hostility		-0.0318* (0.0182)		-0.04300** (0.0182)	-0.0248 (0.0169)	-0.0229 (0.0167)	-0.0236 (0.0176)	-0.0253 (0.0173)	-0.0348* (0.0196)
SharesAcquired		0.0803*** (0.0218)		0.0716*** (0.0226)	0.0678*** (0.0180)	0.0398** (0.0181)	0.0518** (0.0208)	0.0530** (0.0206)	0.0265 (0.0229)
Management		-0.0352** (0.0137)		-0.0326** (0.0143)	-0.0276** (0.0128)	-0.0393*** (0.0126)	-0.0308** (0.0129)	-0.0383*** (0.0128)	-0.0267 (0.0165)
Corruption		0.0308 (0.0203)		0.0203 (0.0215)	-0.0297*** (0.0064)	-0.0301*** (0.0070)	0.0880*** (0.0151)	0.0313* (0.0184)	0.0438* (0.0261)
PoliticalStability		0.0414** (0.0161)		0.0454*** (0.0169)	0.0292*** (0.0068)	0.0009 (0.0082)	0.0690*** (0.0084)	0.0390*** (0.0145)	0.0396* (0.0229)
MarketCaptoGDP		0.0098 (0.0150)		0.0152 (0.0173)	0.0203*** (0.0056)	0.0178*** (0.0061)	0.0085 (0.0109)	0.0051 (0.0143)	0.0011 (0.0159)
_cons	0.4148** (0.1725)	-0.1656*** (0.0482)	0.4432** (0.2080)	-0.1062** (0.0504)	0.0101 (0.0204)	0.0771*** (0.0257)	-0.1721*** (0.0361)	-0.0656 (0.0435)	-0.0885* (0.0534)
Observations	16,223	11,795	16,223	11,795	11,795	11,795	11,795	11,792	5,877
R-squared	0.0389	0.0661	0.0512	0.0788	0.0300	0.0598	0.0526	0.0720	0.0791
Target Nation FEs	YES	YES	YES	YES	NO	NO	YES	YES	YES
Year FEs	YES	YES	YES	YES	NO	YES	NO	YES	YES

Robust standard errors are in parentheses.

† Regression (8) does not include deals for which the Target Nation has less than 10 observations. †† Regression (9) does not include deals for which the Target Nation is the United States.

*** $p < .01$, ** $p < .05$, * $p < .1$

Calculation of BHAR³

The buy-hold abnormal returns (BHAR), used as an alternative measure of run-up, are collected from the WRDS event study tool which uses the following calculation:

$$BHAR_{jt} = \prod_{t=1}^t (1 + R_{jt}) - \prod_{t=1}^t (1 + R_{Market\ Adjusted\ Model})$$

Where R_{jt} is the return for company j at time t , and $R_{Market\ Adjusted\ Model}$ is the expected return based on a market-adjusted model, as shown in the calculation of CAR in the Methodology. For the BHAR, the same estimation windows are used as for CAR.

³ Sourced from Wharton Research Data Services (n.d.)

Table A4 Hypothesis 2 robustness tests, substituting Net National Income for GDP.

	(1) CAR	(2) CAR	(3) CAR	(4) CAR
Targ_NetNIperCapita	0.0027** (0.0012)			
Targ_LogNetNI		0.0747** (0.0351)		
Acq_NetNIperCapita			-0.0011*** (0.0004)	
Acq_LogNetNI				-0.0037 (0.0031)
LogTotalAssets	-0.0115*** (0.0027)	-0.0110*** (0.0027)	-0.0113*** (0.0027)	-0.0112*** (0.0027)
BooktoMarket	0.0055*** (0.0015)	0.0056*** (0.0015)	0.0060*** (0.0015)	0.0059*** (0.0015)
HighTech	0.0199* (0.0104)	0.0206** (0.0104)	0.0218** (0.0106)	0.0214** (0.0107)
PEBidder	-0.0242* (0.0143)	-0.0242* (0.0143)	-0.0220 (0.0150)	-0.0256* (0.0151)
Borrowings	0.0421*** (0.0118)	0.0424*** (0.0118)	0.0442*** (0.0121)	0.0437*** (0.0121)
PercentageofCash	0.0370*** (0.0127)	0.0370*** (0.0127)	0.0387*** (0.0129)	0.0382*** (0.0129)
NumberofBidders	0.1090*** (0.0181)	0.1088*** (0.0181)	0.1031*** (0.0183)	0.1034*** (0.0183)
Hostility	-0.0565** (0.0239)	-0.0570** (0.0238)	-0.0610** (0.0243)	-0.0580** (0.0242)
SharesAcquired	0.1155*** (0.0381)	0.1100*** (0.0381)	0.1119*** (0.0388)	0.1129*** (0.0390)
Management	-0.0230 (0.0481)	-0.0224 (0.0482)	-0.0224 (0.0520)	-0.0208 (0.0523)
Corruption	0.0954** (0.0374)	0.0758** (0.0387)	0.0880** (0.0379)	0.0884** (0.0378)
PoliticalStability	0.0571** (0.0244)	0.0430* (0.0251)	0.0416* (0.0241)	0.0439* (0.0241)
MarketCaptoGDP	0.0045 (0.0200)	0.0038 (0.0192)	-0.0048 (0.0214)	-0.0041 (0.0214)
_cons	-0.3074*** (0.0912)	-0.7361*** (0.2384)	-0.1870** (0.0930)	-0.1905** (0.0938)
Observations	3,218	3,219	3,081	3,081
R-squared	0.1193	0.1189	0.1194	0.1173
Target Nation FEs	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES

Robust standard errors are in parentheses

*** $p < .01$, ** $p < .05$, * $p < .1$

Note. OLS regression results using CAR(-60, -1) as the dependent variable. Intended as a robustness test for the Hypothesis 2 analyses

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